

BEYOND THE CONTEXT WINDOW:
ANCHORING AGENTIC
REASONING WITH WORLD-STATE
AND CONTEXT GRAPHS

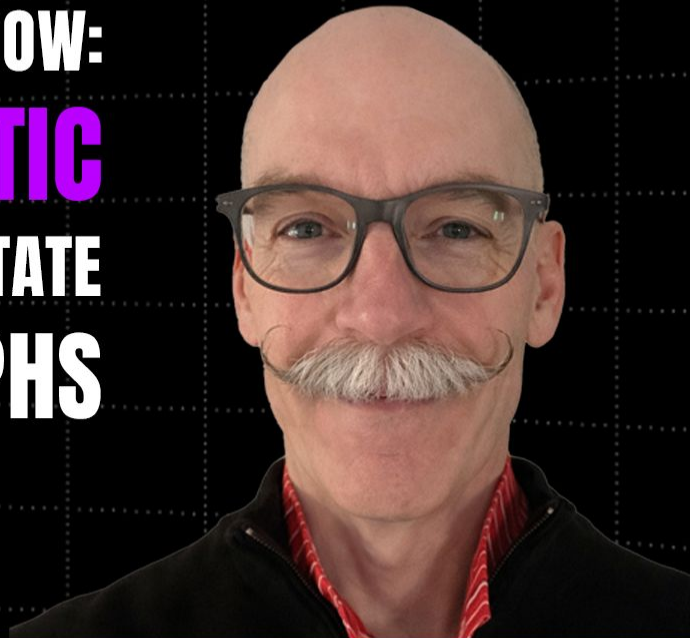


AMY HODLER

Executive Director



GraphGeeks



DAVID HUGHES

Principal Data & AI Solution Architect



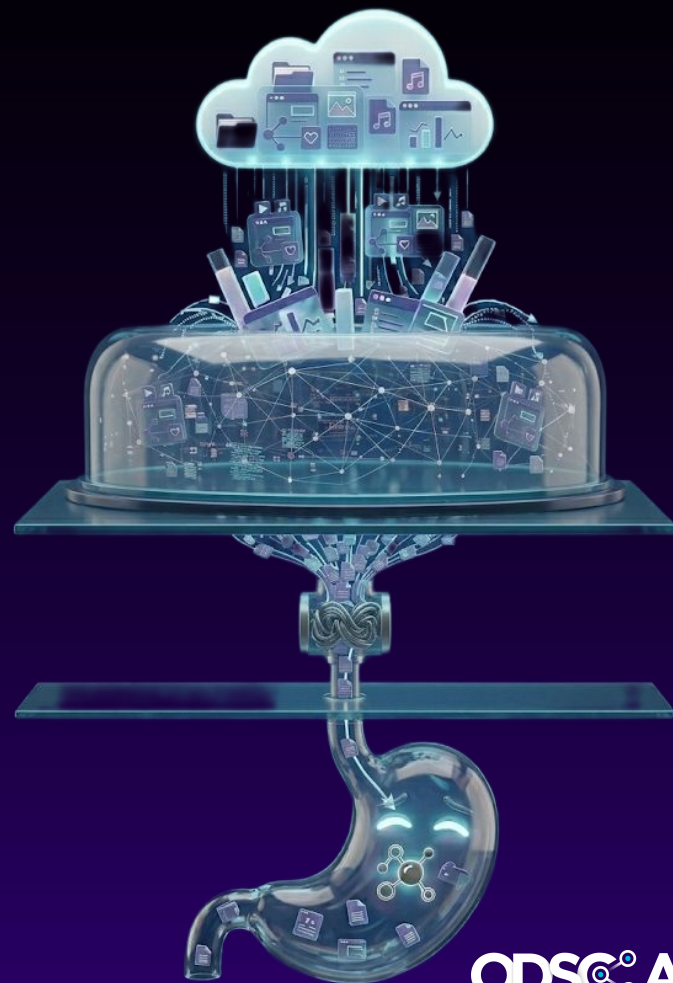
ENTERPRISE
KNOWLEDGE

The Context Window Paradox

Bigger Isn't Always Better

- Tokens Add Up 📦
- Amnesia Persists
 - episodic and procedural
- Multimodal Opacity
- What vs Why

We don't need a bigger stomach we need better digestion

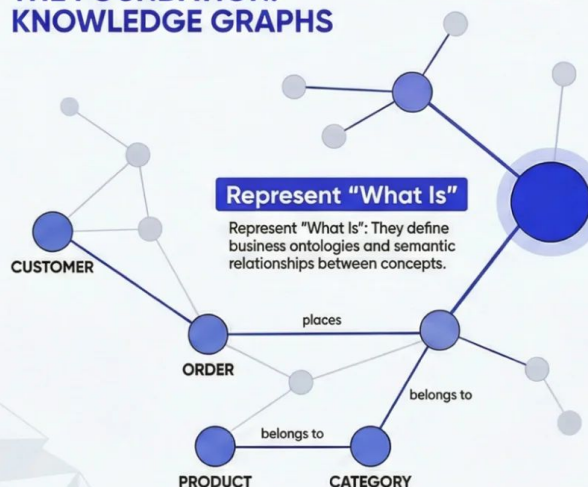


Context Graphs to the Rescue?

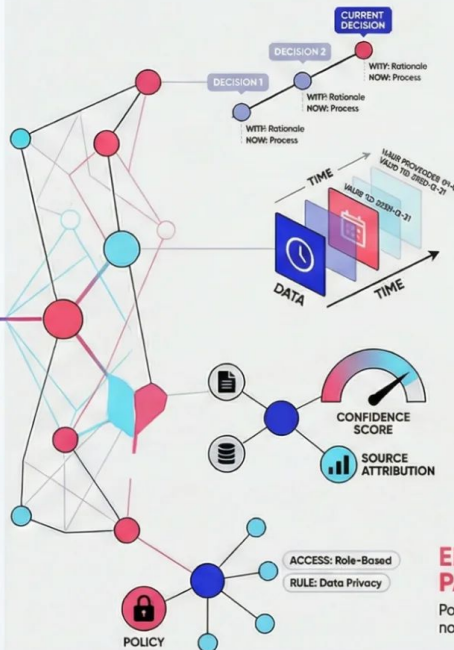
Image via Atlan
Definition via Gartner

THE FOUNDATION: KNOWLEDGE GRAPHS

Connect Static Concepts



WHAT CONTEXT GRAPHS ADD



CAPTURE DECISION HISTORY & PRECEDENT

Records the full "how" and "why" of decisions, making them searchable for AI agents.

ENABLE TIME-TRAVEL FOR DATA

Incorporates timestamps and validity periods to understand how data evolves.

SCORE PROVENANCE & CONFIDENCE

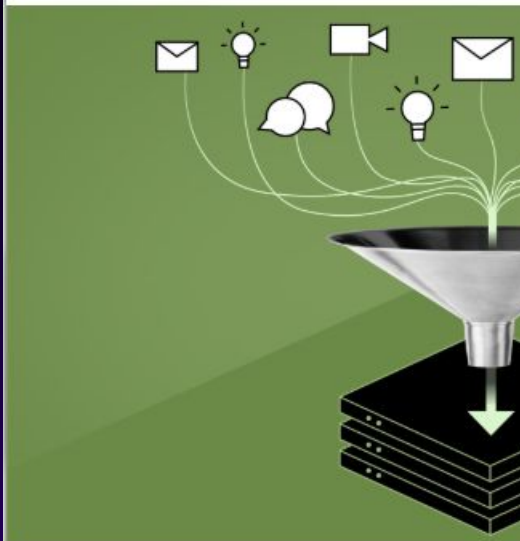
Treats source attribution and quality signals as core parts of the graph.

EMBED GOVERNANCE AS PART OF THE GRAPH

Policies and access rules become queryable nodes, not separate documents.

<https://atlan.com/know/gartner-context-graphs/>

AI's trillion-dollar opportunity: Context graphs



Context Graphs: The Trillion Dollar Elephant

5 min read · Jan 21, 2026



Gaurav Dadhich

Follow



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You can't have a conversation about AI without touching upon Context Graph. Gupta of Foundation Capital put a trillion to the opportunity, AI circles have been 24 hours alone, I have sat through a y conversations about it and read

The Context Graph Hype: Why Engineering Expertise Gets Lost in the Noise

12 min read · Jan 25, 2026



Volodymyr Pavlyshyn

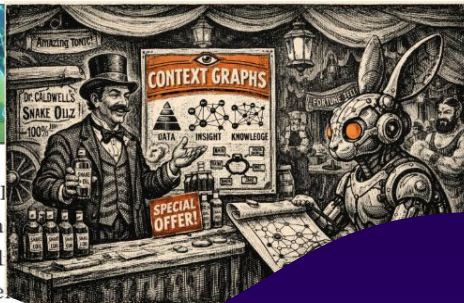
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The Context
Graph
Hype: Why
'Holy Grail' is
Leaking



The Context Graph Hype: Why the "Holy Grail" is Leaking



Todd Blaschka

Advisor | Founder | AI Applications, Blockchain



January 30, 2026

The enterprise

Context Graphs: A Series of Unfortunate Events

Creating Historical Hypergraphs for Event Management



KURT CAGLE

FEB 21, 2026



7



8



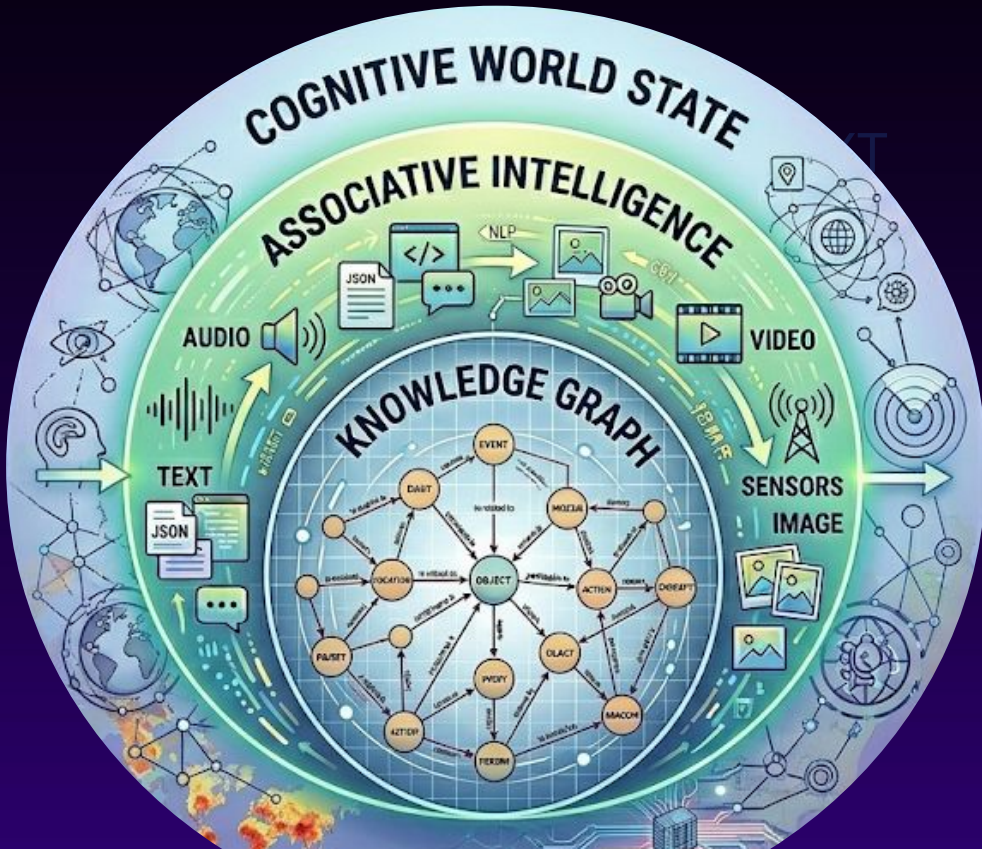
2



No Happy

Context Alone is Not Enough

Agents Need Orientation and Anchoring



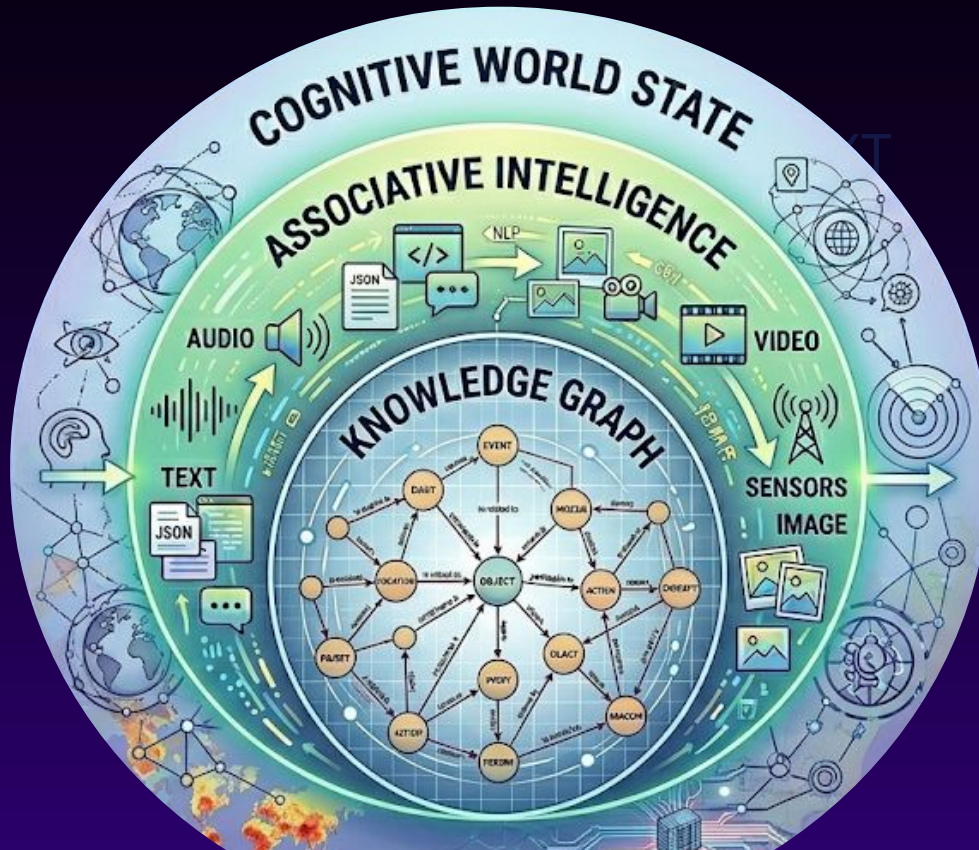
What-If
Simulation

Why
Causal Influence

What Happened
Situational Awareness

What
Facts & Logic

Today: Building Towards a World State



Multimodal & Association

Context + Causal

Blending: Neuro-Symbolic



1

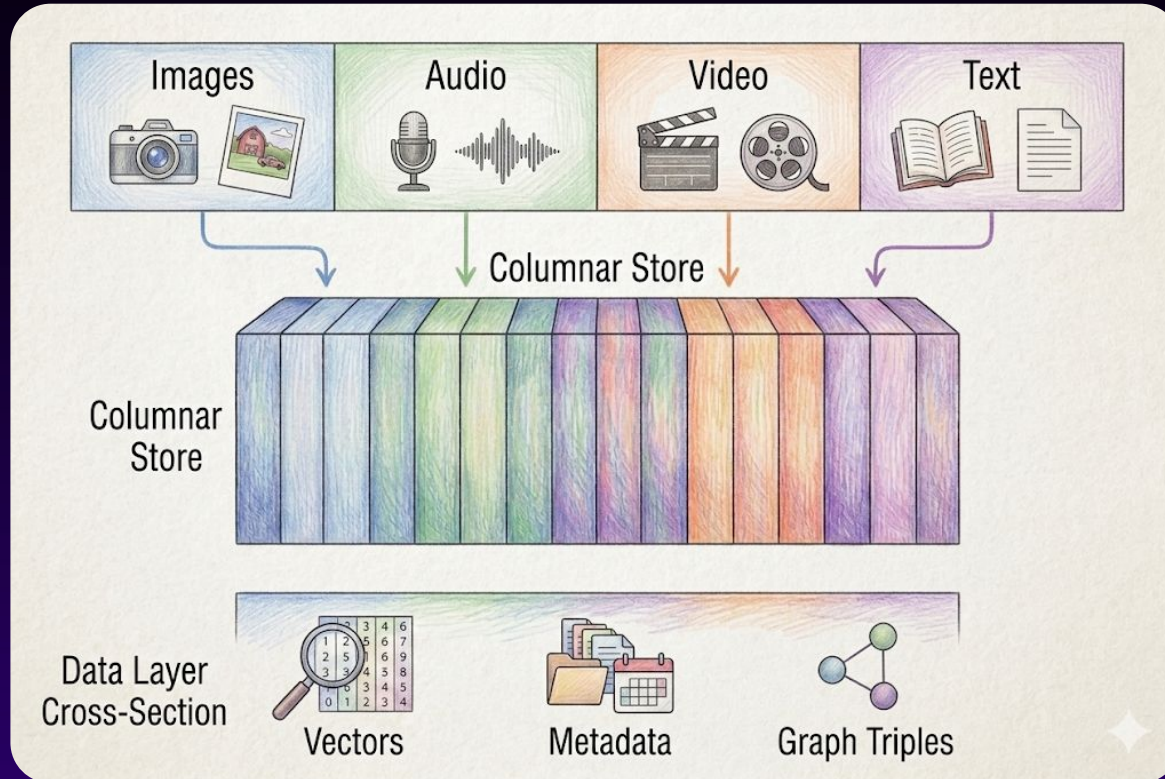
Multimodal Association

We Don't Just Read; We Associate

Real-world complexity isn't solved by one data type, one type of analytics, or one snapshot



Likewise, AI Needs to Connect Complex Data

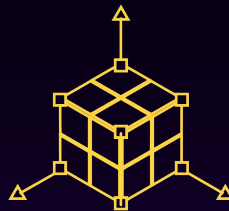


1) "Read" multimodal data

2) Find relationships

3) Act on associations

Multimodal Graph (mmGraph)



Decomposition

Break down
images/audio:
create
subcomponents

Generate
captions

Graph

Components
and subs are
loaded into a
graph

Preserve
context

Embedding Space

Space with
anchors created
for global search
in the graph

Basis of
semantic search

Multimodal Context

Provide blended
results to LLMs
for full context
of text and
images

Associative Intelligence

Connecting Multimodal Data



mmGraph breaks down components that can be explored (texture, spatial placement, etc) individually or together

Blending of semantic context and data (text, visual, etc) enables reasoning across multiple levels of abstraction and associations

An aerial photograph of a naval fleet on the ocean. In the foreground, a large aircraft carrier is visible, moving towards the bottom left. To its right, a smaller ship is moving away, leaving a white wake. Further back, several other ships are visible, including a large cargo ship and a smaller vessel. The ocean is a deep blue, and the sky is clear.

mmGraph High Fidelity Reasoning *within Visual Components*

Are these ships
moving together?

Are there similar
marked ships
nearby?

Contextual
Understanding

Precision Object
Recognition

Nuanced Similarity

Graph Reasoning

Explainability

mmGraph High Fidelity Reasoning *with Layered Audio*



What can't we see?

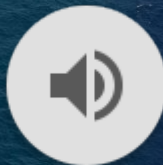
How fast and in
what direction is
that headed?

Adds:

Temporal / Spatial

Directionality & Speed

Off Stage
Object Segmentation

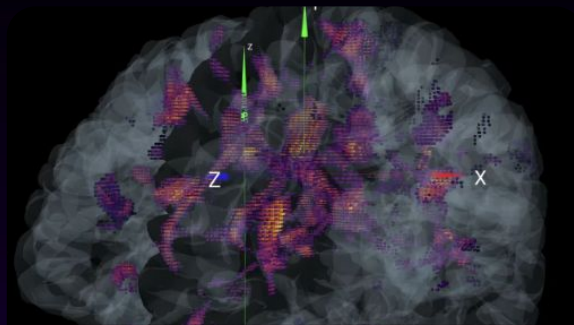




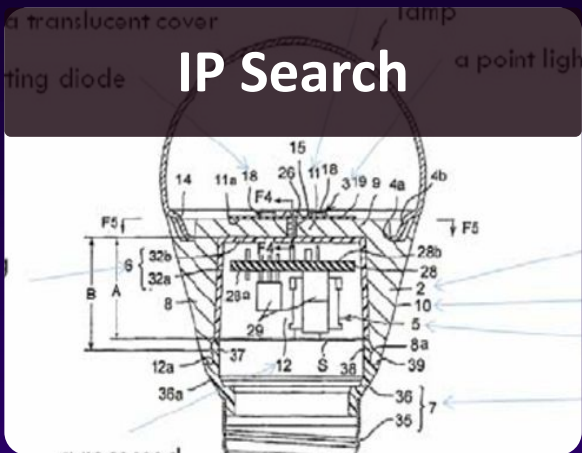
Geospatial Analysis



Security



Medical Diagnostics



IP Search



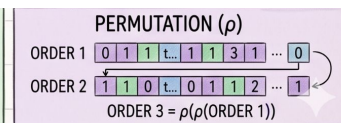
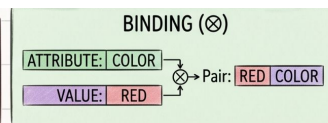
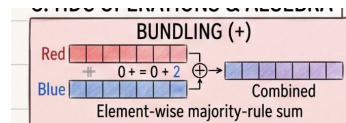
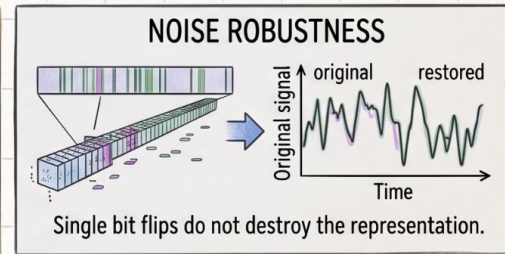
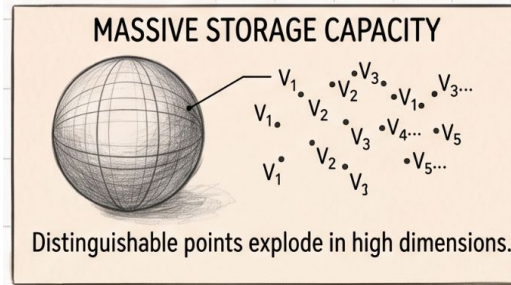
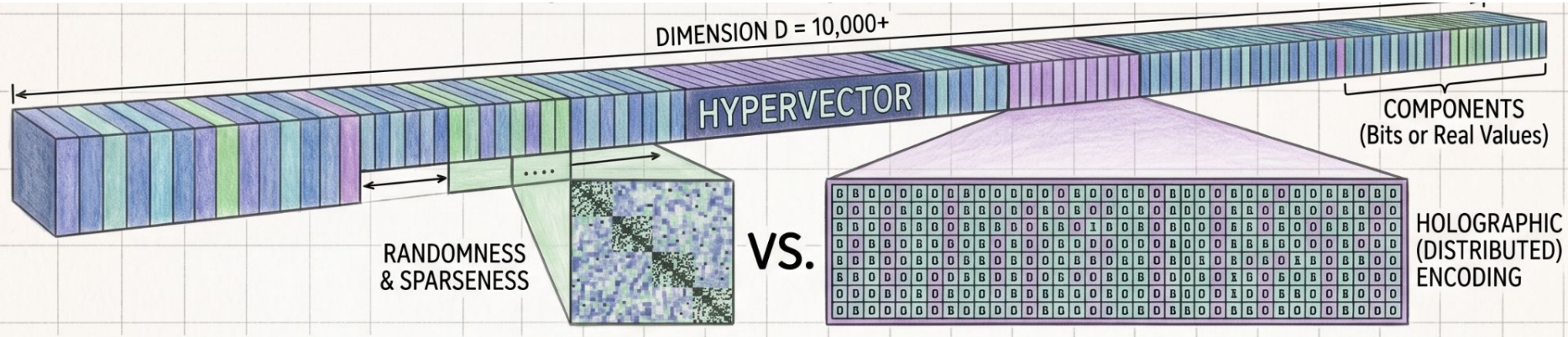
Observation



Predictive Maintenance

Quick Dive into Hypervectors

Hypervector are Super Vectors (10,000+ dimensions)



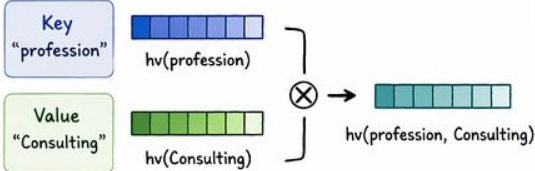
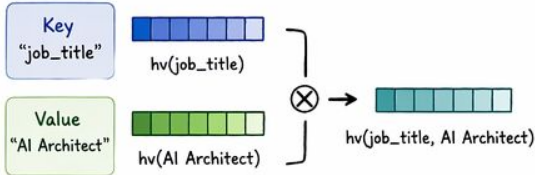
Algebra!

From Path & Properties to a SPO Hypervector

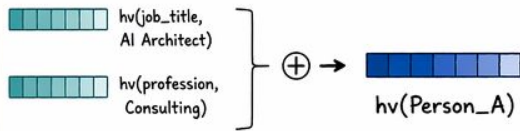
How a graph edge (Person_A — WORKS_WITH → Person_B) is encoded with hypervectors

1) SUBJECT: PERSON_A (hv)

1.1 ENCODE PROPERTIES (KEY & VALUE)



1.2 BUNDLE PROPERTY HVs INTO PERSON NODE HV



Result: Subject (S) hypervector
 $hv(Person_A)$

2) PREDICATE: WORKS_WITH (hv)

2.1 ENCODE PREDICATE

"WORKS_WITH"

$hv(WORKS_WITH)$



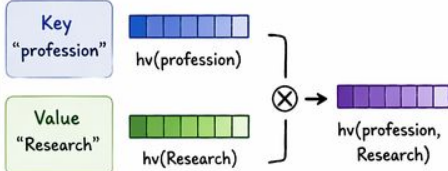
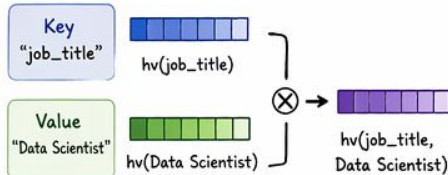
Note:
Predicates typically
have no properties.



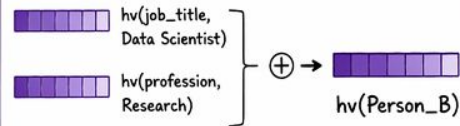
Result: Predicate (P) hypervector
 $hv(WORKS_WITH)$

3) OBJECT: PERSON_B (hv)

3.1 ENCODE PROPERTIES (KEY & VALUE)



3.2 BUNDLE PROPERTY HVs INTO PERSON NODE HV



Result: Object (O) hypervector
 $hv(Person_B)$

4) BIND S, P, O → FINAL SPO

4.1 INPUTS

$S = hv(Person_A)$

$P = hv(WORKS_WITH)$

$O = hv(Person_B)$

4.2 BIND (ASSOCIATIVE BINDING)

$$(S \otimes P) \otimes O$$



Final SPO hypervector
 $hv(SPO)$



Encodes the triple:
(Person_A, WORKS_WITH, Person_B)
with all node properties
embedded via hypervectors.

The background of the slide is a dark blue field filled with a complex, glowing network of light blue and purple lines. These lines, representing circuit traces, are interspersed with various geometric shapes such as squares, circles, and triangles, some of which are also illuminated with a soft glow. The overall effect is a sense of depth and technological complexity.

2

What Happened + Why



CONTEXT GRAPH

- Decision Traces: Map the path from prompt to action
- Tool-Call Histories: Audit external API interactions.
- Policy Enforcement: Validate guardrails were active during inference



CAUSAL ANALYSIS

- Structural Reporting: Retroactive causal tracing (why did this happen).
- Causal Inference: Predicting downstream effects of decisions.
- Testing options with simulated changes

Learn More about Causal ML at www.pywhy.org/



CONTEXT GRAPH



CAUSAL ANALYSIS

- Procedural Record:
Emergent record of operations
 - Decision Log → Holistic:
Often agent only But
Can be Organizational
 - Agentic Grounding &
Continuous Learning
- Analyzes Why:
What caused X to happen?
 - Directional Backbone:
*Calculates influences
on a specific outcome
 - Continuous Learning &
What-If

Use Case Data

International database of sanctions, watchlists, and politically exposed persons and companies

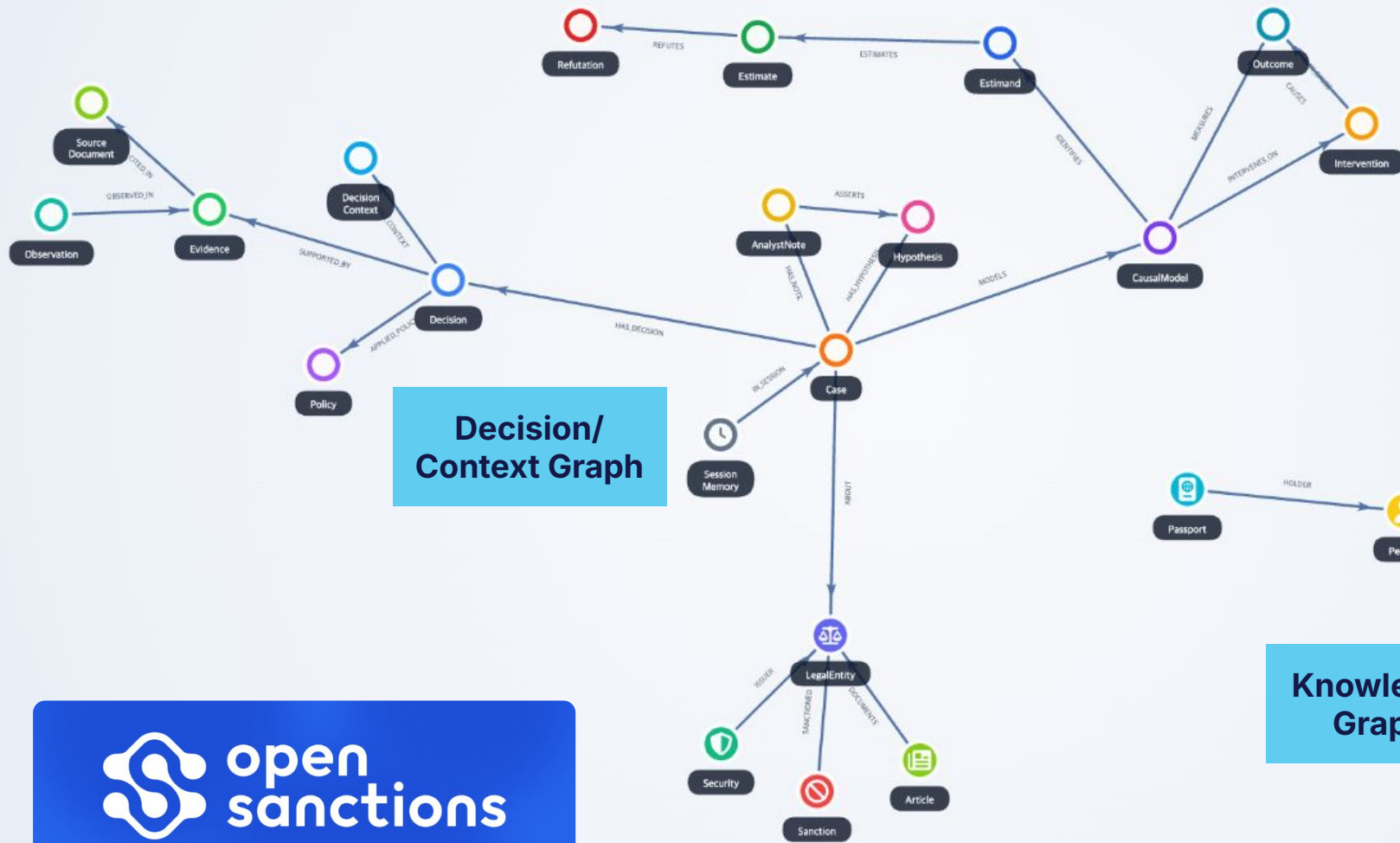
- Several 100 data sources
- Carefully cleaned +ER +HITL
- Bulk access
- Auditable

Open Source with 3 License Options

www.opensanctions.org/



Topics	Entity types
PEP 652,685	People 1,115,372
Politician 254,772	Securities 366,216
Debarred entity 215,802	Companies 237,722
Municipal govern... 81,468	Positions 174,061
Sanction-linked en... 79,346	Organizations 101,638
Close Associate 71,559	Legal entities 83,322
Sanctioned entity 70,352	Addresses 29,932
Wanted 65,966	Cryptocurrency wa... 13,680
Crime 61,971	Vessels 9,609
Executive branch o... 53,816	Articles
Show all...	



Causal Graph

*Some
Sythetic Data

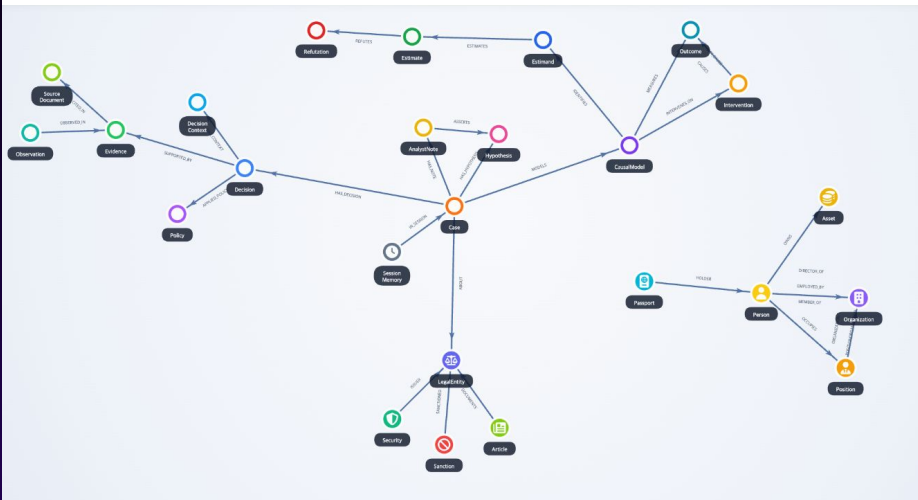




Scenario: Using Context & Causal Analysis *Together*

“Hermes” Investigator:
Does this entity look
suspicious?

- Preserve Reasoning Trail
- Compare Prior Cases
- Ask What-If Questions



HDC Neuromorphic Intelligence
Neurosymbolic Analytics Workbench

← Framework by David Hughes



Context Graph

OpenSanctions Context Graph

Create and inspect durable decision traces, analyst notes, hypotheses, and precedent matches without leaving the workflow workspace.

OPENSANCTIONS Context Workbench

Separate setup, decision traces, and causal testing.

Foundation
Installed

Decision Traces
8 recent cases

Causal Models
1 proposals

PHASE A
Foundation
ready

PHASE B
Decision Traces
ready

PHASE C
Precedents
search pending

PHASE D
Causal Models
ready

Context Network View

10 nodes

10 relationships

Context Graph Workflow

Phase B adds a durable decision-trace workflow for **open_sanctions**. Create a seeded case for a specific LegalEntity id, or leave the field blank to let Hermes pick one with existing graph context.

CONTEXT TRACE
OpenSanctions review: Linkers Automation (Pvt) Ltd.

sanctions_review ·
open · high priority ·
opened 2026-04-
09T04:18:07.564530Z

1 decision

1 evidence

1 note

1 hypothesis

Recent Cases

OpenSanctions review: Genesis Market
Genesis Market · high priority

OpenSanctions review: Genesis Market
Genesis Market · high priority

OpenSanctions review: Taiyuan Jinke Semiconductor Technology Co., Ltd.
Taiyuan Jinke Semiconductor Technology Co., Ltd. · high priority

OpenSanctions review: Linkers Automation (Pvt) Ltd.
Linkers Automation (Pvt) Ltd. · high priority

OpenSanctions review: POSNER, ROBERT S.
POSNER, ROBERT S. · high priority

OpenSanctions review: Beijing Shuohe Zhongkexin Electronic Equipment Co., Ltd.
Beijing Shuohe Zhongkexin Electronic

OPENSANCTIONS
Context Workbench

Separate setup, decision traces, and causal testing.

Foundation

Installed

Decision Traces

8 recent cases

Causal Models

1 proposals

PHASE A

Foundation

ready

PHASE B

Decision
Traces

ready

PHASE C

Precedents

search pending

PHASE D

Causal
Models

ready

Context Network View

11 nodes

11 relationships

Derived from the current query, selected entity, decision trace, precedents, and causal proposal.



Query: 1 Case: 1 Entity: 1 Decision: 1 Context: 1 Policy: 1

Evidence: 1 Document: 1 Observation: 1 Note: 1 Hypothesis: 1

Create Trace

LegalEntity id

Optional NK...

Create Sample Decision Trace

Recent Cases

OpenSanctions review: Genesis Market

Genesis Market · high priority

OpenSanctions review: Genesis Market

Genesis Market · high priority

OpenSanctions review: Taiyuan Jinke Semiconductor Technology Co., Ltd.

Taiyuan Jinke Semiconductor Technology Co., Ltd. · high priority

OpenSanctions review: Linkers Automation (Pvt) Ltd.

Linkers Automation (Pvt) Ltd. · high priority

OpenSanctions review: POSNER, ROBERT S.

POSNER, ROBERT S. · high priority

OpenSanctions review: Beijing Shuoke Zhongkexin Electronic Equipment Co., Ltd.

Beijing Shuoke Zhongkexin Electronic Equipment Co., Ltd. · high priority

Properties

Expand

:

Close

scope context_graph

id case-96140fdebcb7

label Case

name OpenSanctions review: Linkers Automation (Pvt) Ltd.

network_group case

display_name OpenSanctions review: Linkers Automation (Pvt) Ltd.

case_type sanctions_review

title OpenSanctions review: Linkers Automation (Pvt) Ltd.

status open

priority high

opened_at 2026-04-09T04:18:07.564530Z

closed at

PHASE A
Foundation
ready

PHASE B
Decision Traces
ready

PHASE C
Precedents
search pending

PHASE D
Causal Models
ready

Context Network View

12 nodes12 relationships

Context Graph Workflow

Phase B adds a durable decision-trace workflow for **open_sanctions**. Create a seeded case for a specific LegalEntity id, or leave the field blank to let Hermes pick one with existing graph context.

CONTEXT TRACE

OpenSanctions review: Linkers Automation (Pvt) Ltd.

sanctions_review · open · high priority · opened

2026-04-09T04:18:07.564530Z

SUBJECT

Linkers Automation (Pvt) Ltd. · unknown jurisdiction

DECISION PATH

● escalate Confidence 81%

Linked sanction record from ["Department of the Commerce - International Trade Administration", "Entity List (EL) - Bureau of Industry and Security"].

Context: Seeded from OpenSanctions entity Linkers Automation (Pvt) Ltd. with 1 sar links, 0 document links, and 0 security links. · risk high

Policy: Sanctions exposure triage

EVIDENCE

Linked sanction record from ["Department of the Commerce - International Trade Administration", "Entity List (EL) - Bureau of Industry and Security"].

Documents: OpenSanctions entity profile for Linkers Automation (Pvt) Ltd.

Observations: Linkers Automation (Pvt) Ltd. (NK-2026-04-09T04:18:07.564530Z) has 1 sanction edges, 0 document edges, and 0 in security graph.

Decision Trace

- Preserves not only outcome but part of the why
- Session memory & analyst notes are reusable reasoning assets (often lost after interference)

SESSION MEMORY

Session memory: Created a seeded OpenSanctions decision trace for Linkers Automation (Pvt) Ltd..

ANALYST NOTES

Hermes Analyst · analysis

The entity appears relevant for escalation review because it is a Pakistan-based company in a sanctions-sensitive technology and export-control context. The current trace should be treated as a compliance screening lead that requires provenance review, ownership review, and confirmation of any direct sanctions linkage.

Hypotheses: Companies operating in export-sensitive industrial or automation sectors in higher-risk jurisdictions are more likely to require escalation when they also show sanctions-adjacent evidence or connected-source provenance.

Precedents

Run precedent search after you have a loaded case trace and at least a few stored cases.

Comparison Summary

Use as Causal Population

Load Precedent Trace

Clear Comparison

OpenSanctions review: Linkers Automation (Pvt) Ltd. (Linkers Automation (Pvt) Ltd.) vs OpenSanctions review: Taiyuan Jinke Semiconductor Technology Co., Ltd. (Taiyuan Jinke Semiconductor Technology Co., Ltd.) - Similarity 1.026

Both cases share decision outcomes: escalate. They align on risk levels: high. Both rely on similar evidence types: open_sanctions_entity_snapshot. They reference overlapping policies: Sanctions exposure triage. Their provenance comes through overlapping publishers: open_sanctions. The main distinctions are in the language used across notes, decisions, and evidence.

Analyst Guidance

Strengths

- The precedent is strong on decision outcome and risk alignment.
- The cases are supported by similar evidence types.
- The same policy framing appears in both cases.
- The comparison includes overlapping provenance rather than text overlap alone.

Recommended Review

- Review the source-case-only documents before relying on the precedent.
- Read the analyst notes and decision summaries directly to understand the language differences.

Cautions

- Unique documents on each side may impact investigative conclusion.

Shared Signals

Precedent Comparison

- Shared signals & Key Differences
- Analysts reasoning aid, not just a score

Shared Signals

Outcomes

escalate

Risk Levels

high

Evidence Types

open_sanctions_entity_snapshot

Policies

Sanctions exposure triage

Publishers

open_sanctions

Document Types

entity_profile

Shared Documents

none

Shared Language

administration appears based because bureau
commerce company compliance connected context
control current department entity escalation

Key Differences

Linkers Automation (Pvt) Ltd. publishers

none

Taiyuan Jinke Semiconductor Technology Co., Ltd. publishers

none

Linkers Automation (Pvt) Ltd. documents

OpenSanctions entity profile for Linkers Automation (Pvt) Ltd.

Taiyuan Jinke Semiconductor Technology Co., Ltd. documents

OpenSanctions entity profile for Taiyuan Jinke Semiconductor Technology Co., Ltd.

Linkers Automation (Pvt) Ltd. documents

Proposed causal model for case 'OpenSanctions review: Linkers Automation (Pvt) Ltd.':
intervention 'Enhanced Screening' may affect outcome 'Escalation Decision'.

Causal Graph

CAUSES disabled

**OpenSanctions review:
Linkers Automation (Pvt) Ltd.
causal model**

CausalModel · model · proposed

**OpenSanctions review:
Linkers Automation (Pvt) Ltd.**

Case · case · open

MODELS

OpenSanctions review: Linkers Automation (Pvt) Ltd. → OpenSanctions review: Linkers Automation (Pvt) Ltd. causal model

IDENTIFIES

OpenSanctions review: Linkers Automation (Pvt) Ltd. causal model → risk_difference

risk_difference

Estimand · estimand · not_identified

Enhanced Screening

Intervention · treatment · proposed

INTERVENES_ON

OpenSanctions review: Linkers Automation (Pvt) Ltd. causal model → Enhanced Screening

MEASURES

OpenSanctions review: Linkers Automation (Pvt) Ltd. causal model → Escalation Decision

Escalation Decision

Outcome · outcome · proposed

MAY_CAUSE

Enhanced Screening → Escalation Decision · 25%

MAY_CAUSE

Enhanced Screening → Escalation Decision

Status: proposed_only

Confidence: 25%

Analyst-proposed causal direction pending identification, estimation, and refutation.

Treatments / Interventions

Enhanced Screening · Treatment/intervention variable proposed for OpenSanctions review: Linkers Automation (Pvt) Ltd. causal model.

Outcomes

Escalation Decision · Outcome variable proposed for OpenSanctions review: Linkers Automation (Pvt) Ltd. causal model.

Estimands

risk_difference · not_identified · precedent_matched_cases · 9eae703946b6 · Enhanced Screening → Escalation Decision

Proposed Edges

Enhanced Screening MAY_CAUSE Escalation Decision

Causal Proposal

- Assumptions and providence stay visible
- Intervention, outcome, and estimand are explicit
- Tune guardrails for HITL review, workflow tasks, etc.

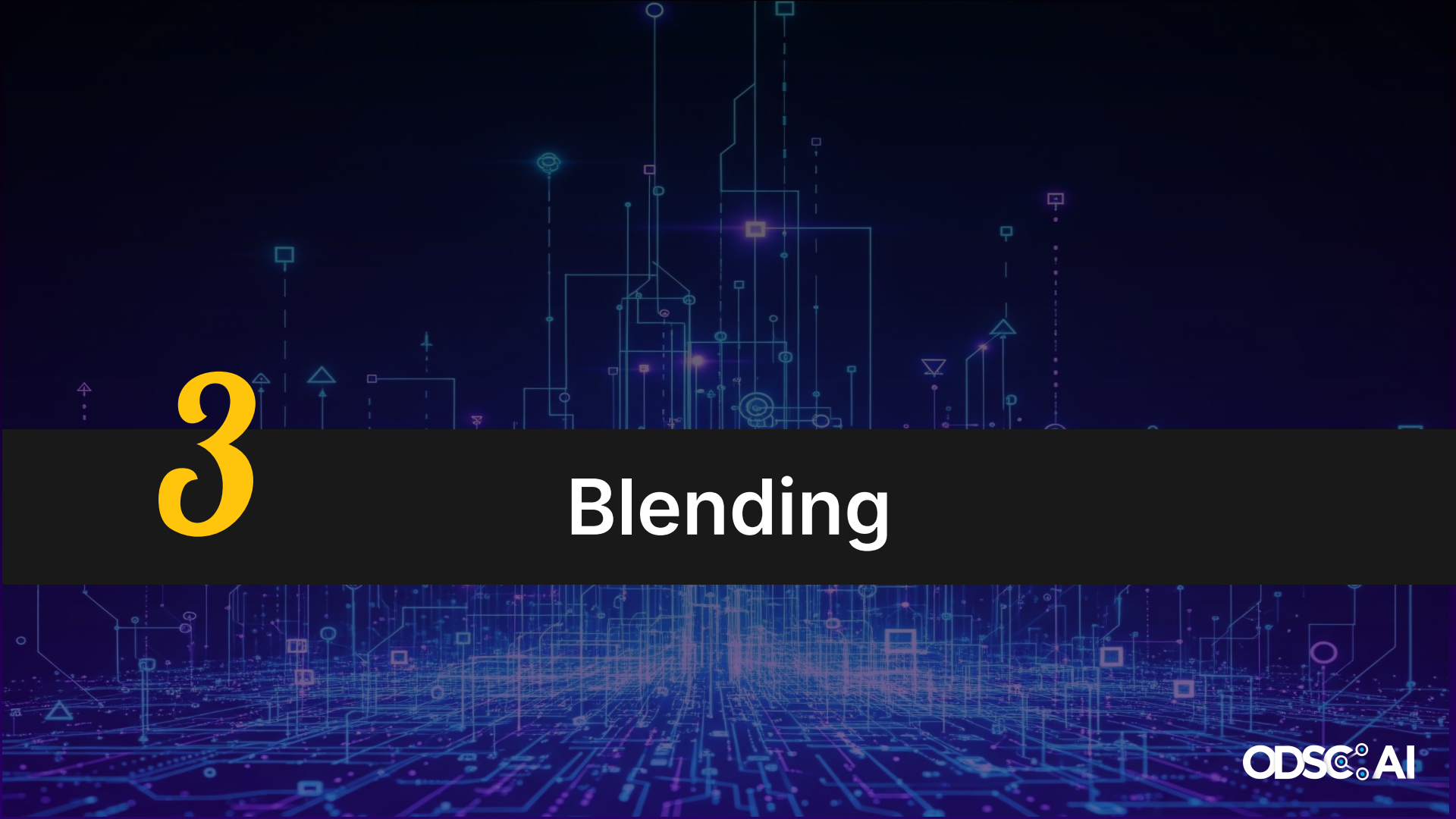
no estimate

no refutation

no accepted CAUSES claim

Guardrails: MAY_CAUSE edges are proposals only. Accepted CAUSE edges require operational outcome data, refutation, and human review. Synthetic estimates are workflow test output, not causal evidence.

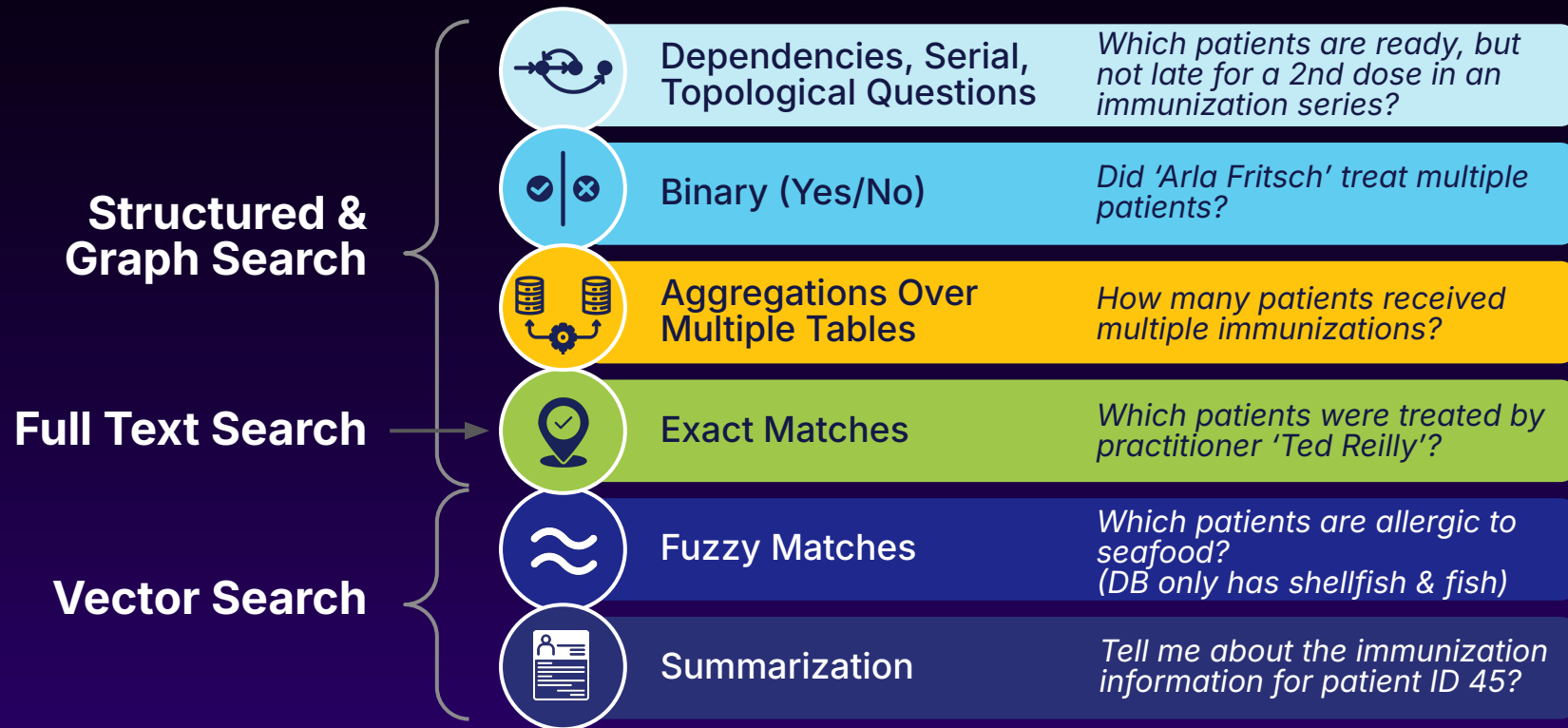
This is a causal model proposal only. No identification, estimation, or refutation has been run.



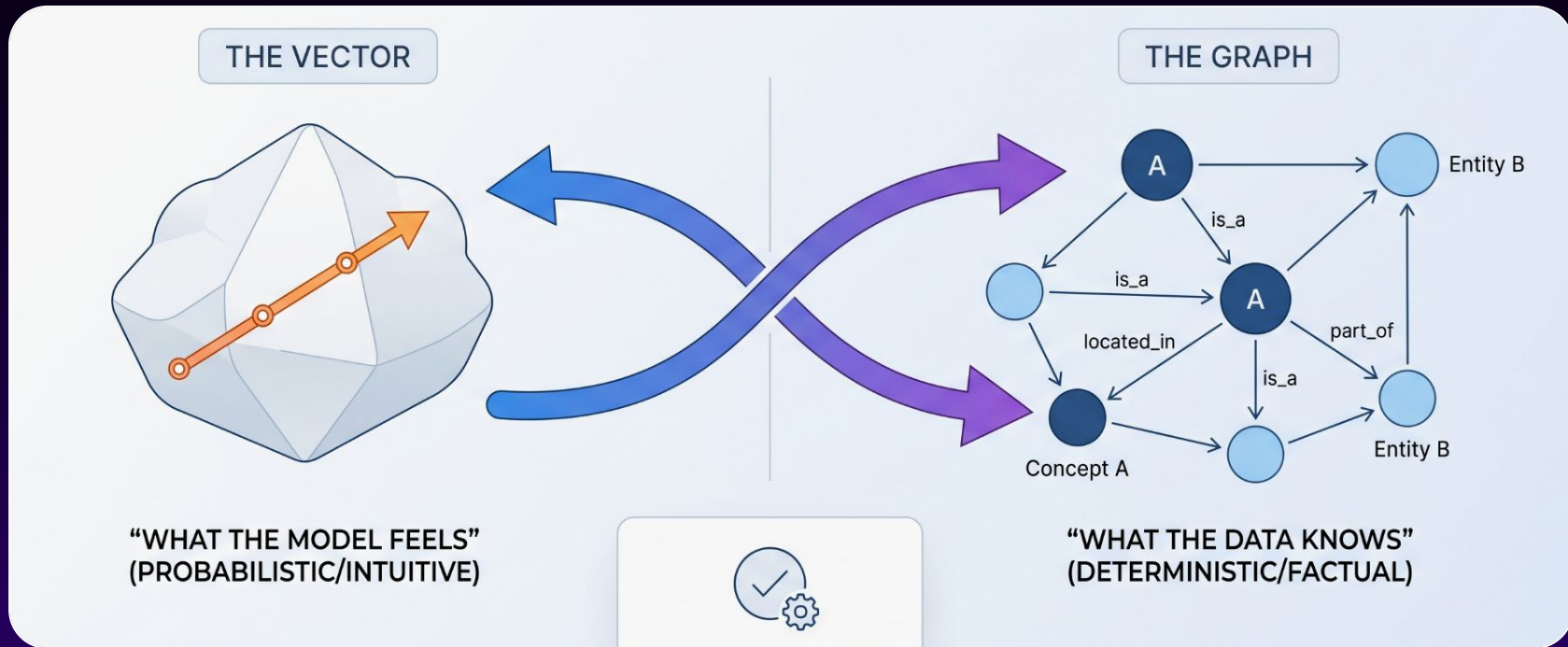
3

Blending

Hybrid: Use the Best Tool for the Job



Can We Blend at the Onset the Intuition of Vectors + the Facts of Graphs?





Triple optional

Top-K

- 3 +

Returned vectors

80

View

2D

3D

Node size

Label size

☒ Labels

SARA JEAN KROL - INVOLVED_IN - JL000260 · 2024-06-29 11:00:00+00:00

Showing 80 triple vectors. Select one to draw top-3 neuro-semantic similarity links. Drag to rotate and scroll to zoom.



Triple Embeddings (SPO Bound Hypervector)

- Similar triples are nearest neighbors
- Foundation of assoc graph traversal

HDC

Expand

:

Close

event_time

2024-12-24 11:00:00+00:00

composition

$\text{normalize}(\text{subject_vector} * \text{relation_key} * \text{object_vector})$

Embedding

10000 dimensions ·

$\text{normalize}(\text{subject_vector} * \text{relation_key} * \text{object_vector})$

[0.0054, 0.0150, -0.0001, -0.0053, -0.0054, -0.0054, -0.0016, -0.0010, -0.0016, -0.0135, -0.0012, -0.0069, ...]

Demo triple similarities

subject_to_object

-0.0045

subject_to_triple

-0.0088

object_to_triple

-0.0183

Top-3 similar triple vectors

KEIFER LATHAM COOPER-INVOLVED_IN-JL000250 (93.5%)

KEIFER LATHAM COOPER-INVOLVED_IN-JL000775 (91.2%)

KEIFER LATHAM COOPER-INVOLVED_IN-JL000447 (64.8%)

HDC Traversal Compare

Compare exact discrete graph paths with associative pure-HDC traversal, then inspect where both agree and where HDC suggests nearby alternatives.

Query family

Person - CDREvent - CellPhoneTower

Preset

balanced

Case

ROBBY JO PERKINS -> JL000241 -> tower tower-012

Run Traversal Compare

☒ Show invalid associative tail

Exact overlap

2

Same-anchor associative

2

Associative-only

2

Neuro-symbolic coherence

50%

Discrete Results

Exact graph-valid paths from the Lance graph/Cypher traversal.

person-00014 -> JL000241 -> tower tower-012
Path id: person:person-00014|event:241|tower:tower-012

person-00014 -> JL000564 -> tower tower-044
Path id: person:person-00014|event:564|tower:tower-044

person-00014 -> JL000002 -> tower tower-073
Path id: person:person-00014|event:2|tower:tower-073

Pure-HDC Raw Top-K

The unranked candidate list before symbolic validation and coherence weighting.

ROBBY JO PERKINS -> JL000241 -> tower tower-012
Raw #1 · score 1.0000

ROBBY JO PERKINS -> JL000564 -> tower tower-044
Raw #2 · score 0.3241

Associative Results

Pure-HDC reranked results with symbolic validation layered on after retrieval.

Exact 2 · Same-anchor 0 · Associative 0 · Invalid 0

ROBBY JO PERKINS -> JL000241 -> tower tower-012 exact_match
Raw #1 -> reranked #1 · score 1.0000
same_anchor · shared_event · shared_tower

ROBBY JO PERKINS -> JL000564 -> tower tower-044 exact_match
Raw #2 -> reranked #2 · score 0.6282
same_anchor · shared_event · shared_tower

ROBERT CARR -> JL000203 -> tower tower-060 schema_valid_only
Raw #3 -> reranked #3 · score 0.1494
near_symbolic_cluster

SAUL ADRIAN ZAPATA -> JL000387 -> tower tower-085 schema_valid_only
Raw #4 -> reranked #4 · score 0.1352
near_symbolic_cluster

Hybrid Results

Graph-valid expansion with HDC scoring for ranking. This is the operational high-confidence lane.

ROBBY JO PERKINS -> JL000241 -> tower tower-012
score 1.0000

Comparison of Different Methods:

- **Discrete Results** tell us which graph paths are valid
- **Pure-HDC Raw Top-K** s shows what's nearby in associative space
- **Associative Results** is the HDC with symbolic validation
- **Hybrid Results** keeps the discrete results but uses an associative ranking of results

HDC Traversal Compare

Compare exact discrete graph paths with associative pure-HDC traversal, then inspect where both agree and where HDC suggests nearby alternatives.

Query family

People connected through a shared tower ▾

Preset

balanced ▾

Case

DAVID ALLEN BRADFIELD -> tower tower-004 -> TIMOTHY MICHAEL DACEY ▾

Run Traversal Compare



Show invalid associative tail

Exact overlap
19

Same-anchor associative
20

Associative-only
1

Neuro-symbolic coherence
95%

Quantify Agreement of Exact & Associative

- **Exact Overlap:** where associative retrieval reproduces exact paths
- **Same-Anchor Assoc:** results that are anchored in a point also on a discrete path (toe in the water)
- **Associative-Only:** where HDC suggests additional nearby results
- **Neuro-symbolic coherence:** summarizes the level of agreement



Manifold Analysis

- **Green** Semantic-only results
- **Orange** Structural-only
- **Bright** intersection region is where the two signals agree

Color by hybrid agreement $H(n) = w_e(n) \cdot w_g(n)$: red = structural-only, green = semantic-only, yellow = intersection, white-hot = highest $H(n)$. Nodes glow by semantic-structural coherence.

Graph node placement: Centroid: graph-only nodes (e.g. PhoneNumber, Location) are placed at the geometric center (average position) of their linked embedding nodes. Keeps nodes among their connections; compact layout.

Shells: ■ Embedding (selected + top-k) ■ Graph (selected + hops) ■ Intersection ($E \cap G$)

Jaccard coherence
4%

Hybrid coherence
84%

Shell separation
1.23
D / avg radius

Semantic coverage
75%
 V_I / V_E

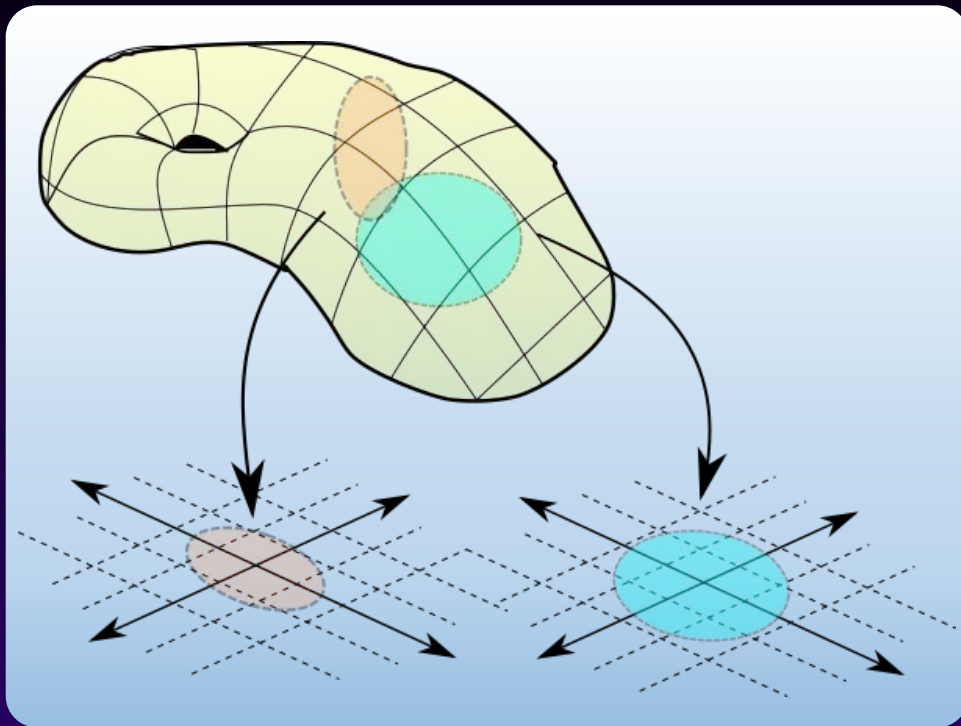
Structural coverage
12%
 V_I / V_G

Hybrid radar



Sem = semantic coverage, Str = structural coverage, Hyb = hybrid coherence

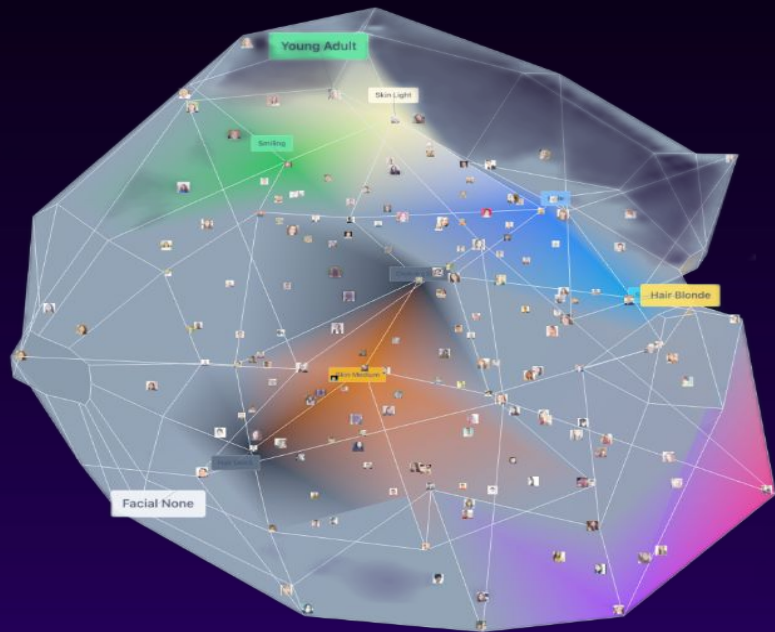
Manifolds Bridge Complexities



A Manifold *locally looks like ordinary flat space* (so we can use common tools), even if its overall, global shape is much more complex.

Often used for visualizing high-dimensional data

Manifold Blending for Neurosymbolic Analysis



- Classification & grouping along any given angle and magnitude (vector space and/or topological space)
- Conceptual representation, not just semantic or structural
- Query sequentially, separately, or together

The background of the slide is a dark blue field filled with a complex, glowing network of light blue and purple lines. These lines, representing circuit traces, are interspersed with various geometric shapes such as squares, circles, and triangles, some of which are also illuminated with a soft glow. The overall effect is a futuristic, high-tech aesthetic that suggests a digital or cognitive environment.

Building a Cognitive World State

1) STEP 1: MULTI-MODAL INGESTION

IMAGE INGEST



Object recognition,
visual features

AUDIO INGEST



Speech-to-Text (STT),
audio features

VIDEO INGEST



Frame sequence analysis,
visual + audio temporal

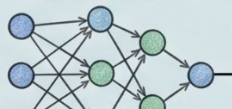
TEXT INGEST



Natural Language Processing (NLP),
semantic analysis

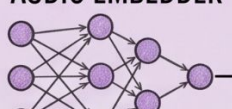
2) STEP 2: EMBEDDING & TRANSFORMATION

VISUAL EMBEDDER



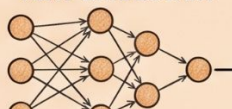
hv(Image) hv(Video)

AUDIO EMBEDDER



hv(Audio)

TEXT EMBEDDER



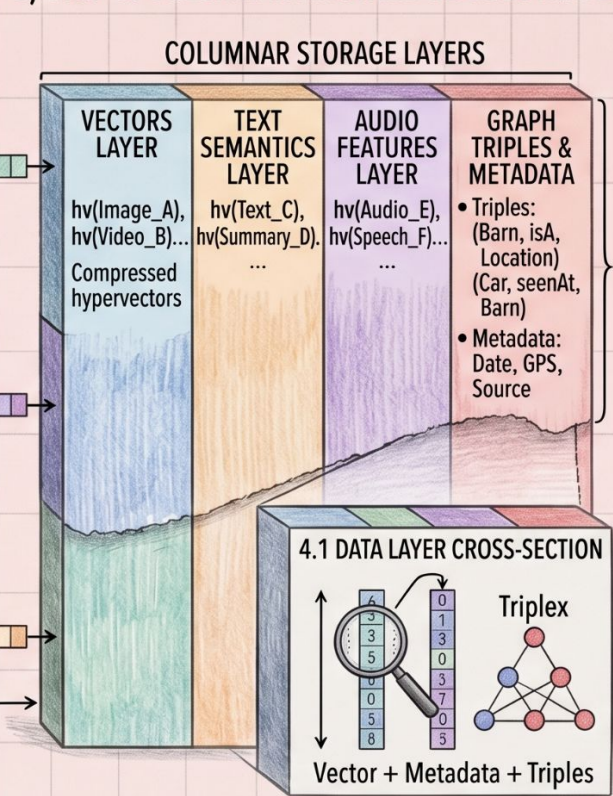
hv(Text)

METADATA & TRIPLEX ENCODER



Subject (S), Predicate (P),
Object (O)

3) STEP 3: COLUMNAR KNOWLEDGE SUBSTRATE



4) STEP 4: APPLICATIONS & REASONING

HYBRID SEARCH



Find similar audio, related
images, and semantic context

CROSS-MODAL REASONING



Form patterns
between
different media,
discover hidden links

KNOWLEDGE GRAPH VIEW



Visualize facts,
explore entity relationships

MULTIMODAL SUMMARIZATION



Create comprehensive
summaries across all inputs

First Unify Multimodal Knowledge & Associate

1) STEP 1: MULTI-MODAL INGESTION

IMAGE INGEST



Object recognition,
visual features

AUDIO INGEST



Speech-to-Text (STT),
audio features

VIDEO INGEST



Frame sequence analysis,
visual + audio temporal

TEXT INGEST



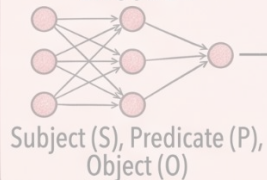
Natural Language Processing (NLP),
semantic analysis

2) STEP 2: EMBEDDING & TRANSFORMATION

VISUAL EMBEDDER

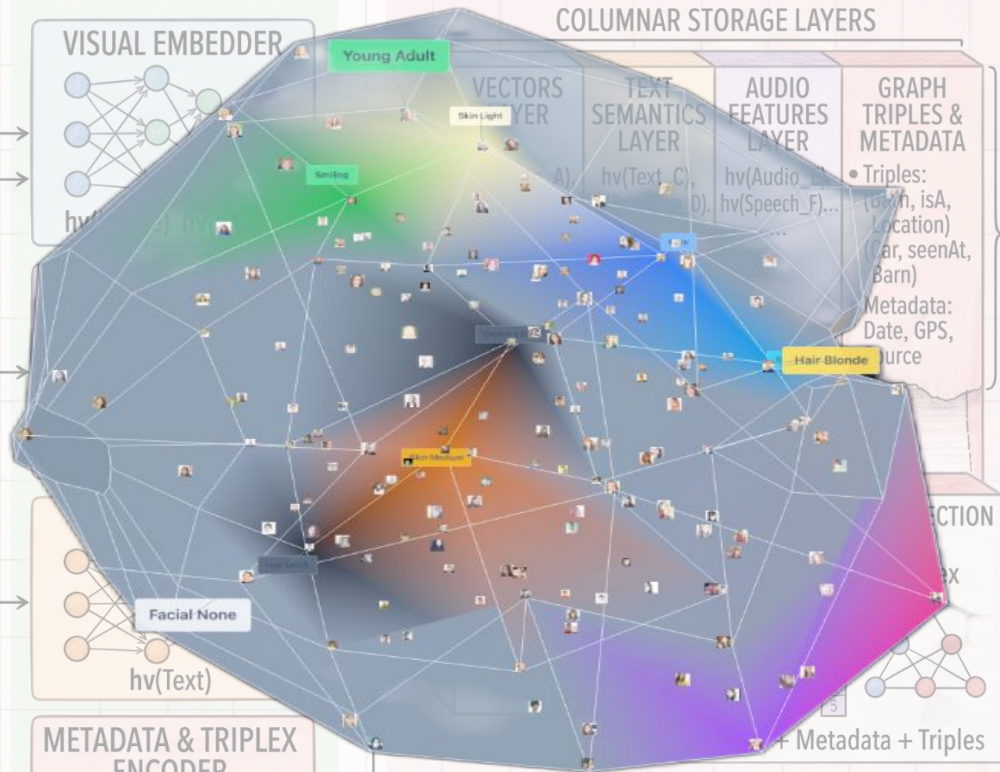


METADATA & TRIPLEX ENCODER



3) STEP 3: COLUMNAR KNOWLEDGE SUBSTRATE

COLUMNAR STORAGE LAYERS



4) STEP 4: APPLICATIONS & REASONING

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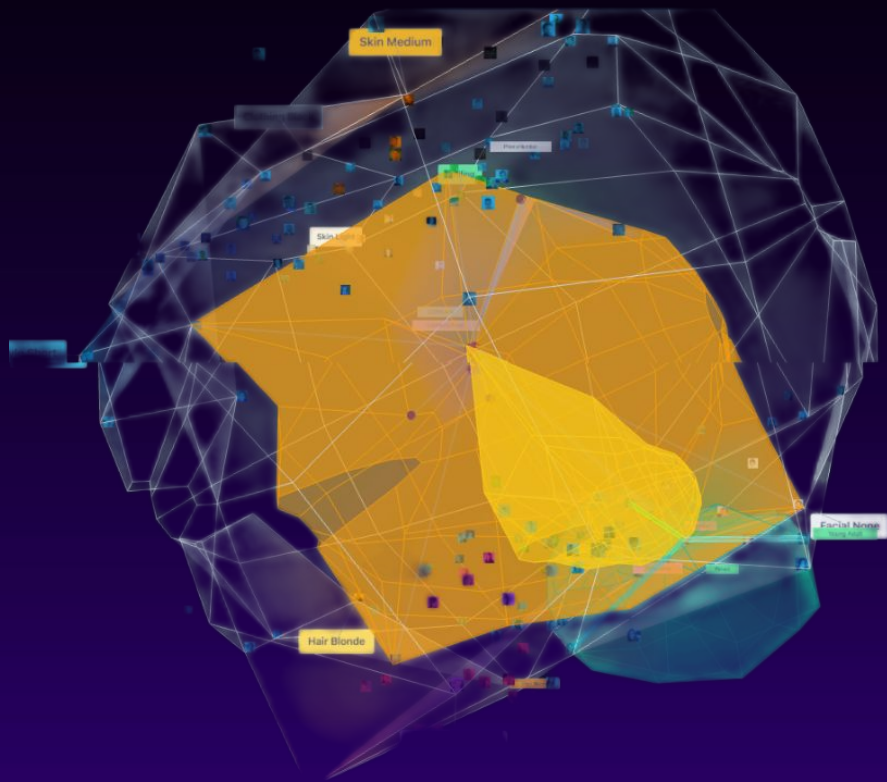
MULTIMODAL SUMMARIZATION



Create comprehensive
summaries across all inputs

Then BLEND

Meaning Emerges not as points, but as Regions

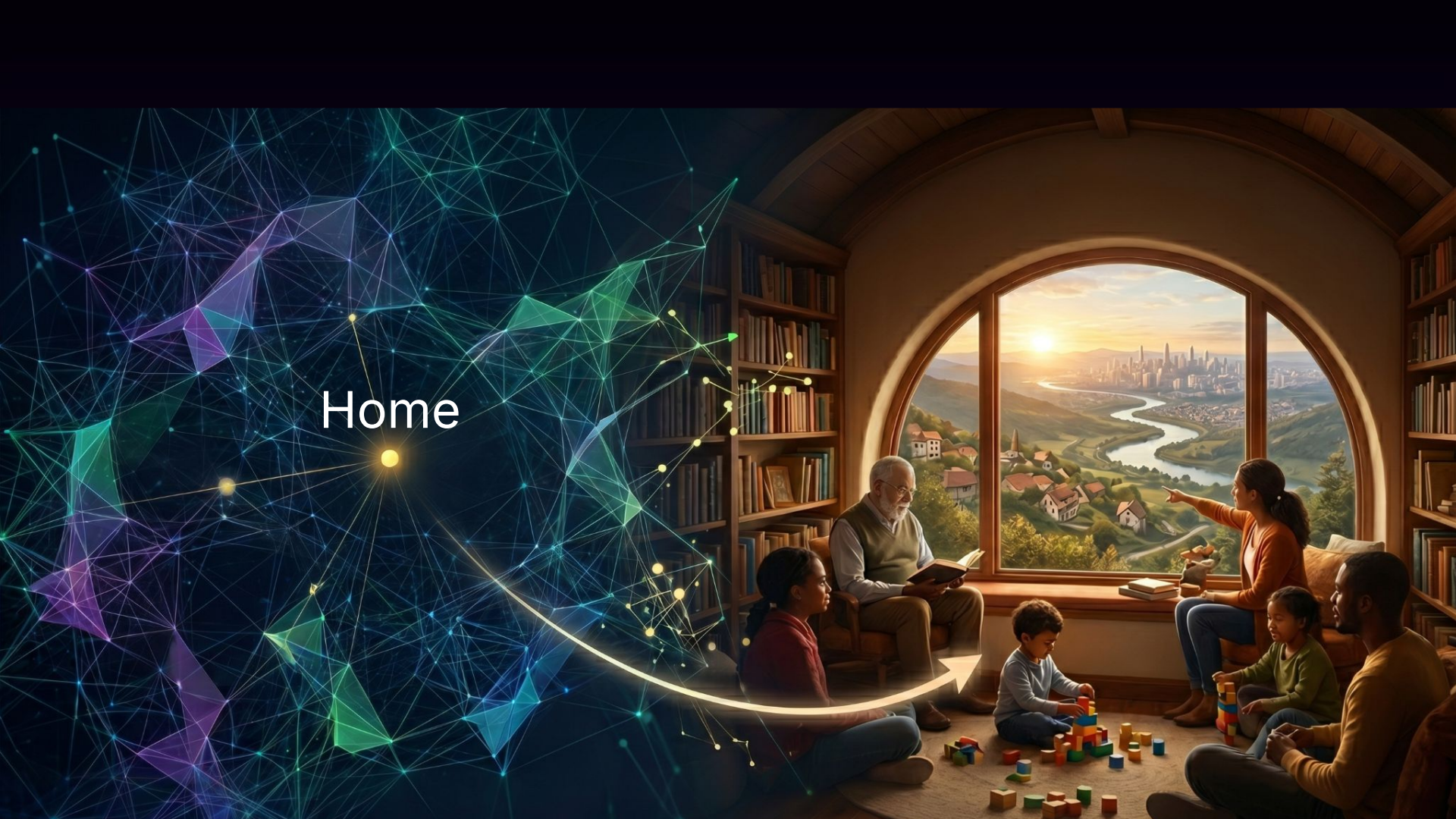


Curse of Dimensionality:
In high-dimensional spaces,
everything is far apart

Folding Reduces that space

Resulting regions are
Cognitive Objects:

- Attracting related concepts
- Resisting noise
- Supporting reasoning & traversal

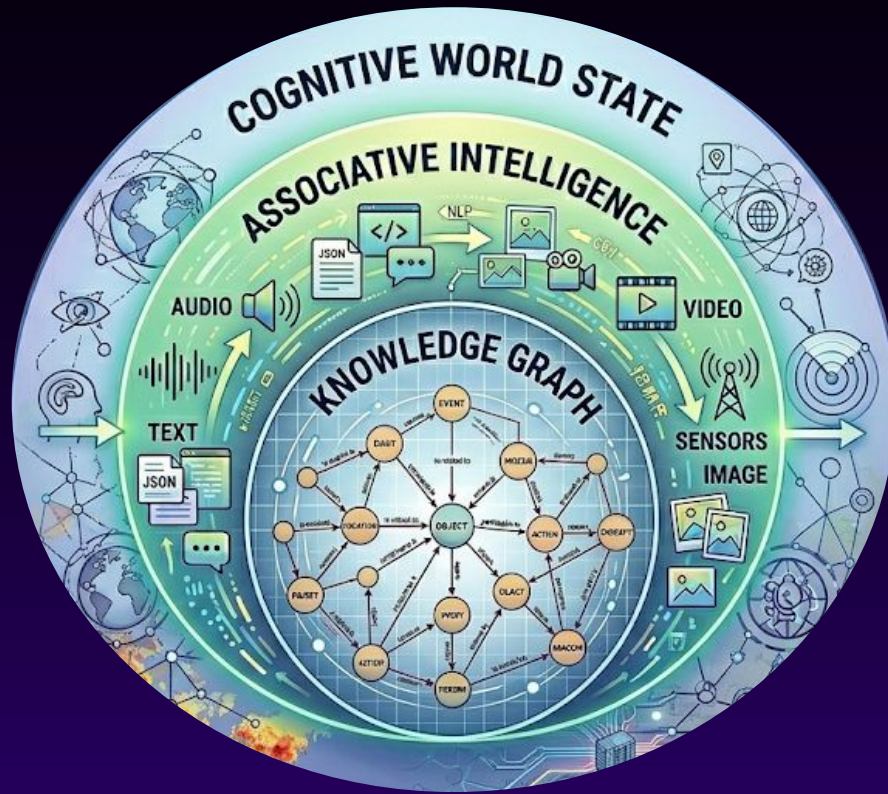
The image is a composite. The right side shows a family of six in a cozy library with a vaulted ceiling and floor-to-ceiling bookshelves. An elderly man with a white beard sits in a chair, reading a book. A woman sits on the floor, pointing towards a large arched window. Three children are on the floor playing with colorful blocks. The window looks out onto a scenic landscape with a winding river, a small village, and a distant city skyline under a sunset sky. The left side of the image is a dark blue space filled with a complex, glowing digital network of lines and nodes. A bright yellow light source is at the center of this network, and a glowing white arc with an arrow points from it towards the family in the library.

Home

Blend Perspectives for a Cognitive Space Both AI and Humans Can Use

Home





Anchoring AI in neuro-symbolic architectures, ensures a shared cognitive logic that prevents divergence;

AI can progress at speed while remaining grounded in human-centric reasoning

<https://github.com/stugorf/odsc-east-2026>



**Amy
Hodler**



**David
Hughes**



GraphGeeks.org
for the love of connecting

ODSC AI

Other Graphy In-Person Sessions:

The Compounding Power of Connected Data (today/tomorrow) - 30 mins

Entity Resolved Knowledge Graphs:
Effective GraphRAG - Today: 11-1pm Rm 310

The Ontology Layer Your Data Pipeline Is Missing - Today: 2-3pm Rm 310

Book: Practitioner's Guide to Graph Data -
Today 3pm Expo

Leveraging Graphs & Ontology for Better
Context Engineering -
Today 3:30-5:30pm Rm 310

tiny.graphgeeks.org/mixer

Graphy Socializing

